

# Constraint Is Not Geometry

**Subtitle.** A preregistered counterexample to reachability-aligned causal deformation in a frozen meta-recurrent agent

**Author.** Jawaun Brown

**Date.** 27 July 2026

**Status.** Agent-generated experiment and manuscript draft under human review

## Abstract

Does changing the rule that determines successful futures necessarily deform an agent's internal similarity geometry toward those futures, and does that deformation cause the changed behavior? We tested this claim in a finite world where every future-language element was exactly enumerable. Thirty-two paired meta-GRUs inferred one of two hidden, balanced future-admissibility constraints from matched demonstrations. Observations, transition dynamics, feedback channel, timing, magnitude, marginal label entropy, architecture, initialization scheme, and training budget were controlled. Both tasks and a learnable control were solved perfectly (mean accuracy 1.000); a randomized-label sham remained at chance (0.509); and an injected-geometry positive control was recovered strongly (alignment lift 0.917). The registered constraint-specific geometry gate nevertheless failed in both directions: active-minus-inactive partial alignment was -0.363 for constraint A and -0.237 for B. Counterbalanced swaps moved the geometry opposite the prediction, and rank-, norm-, and covariance-matched latent transports failed selective impairment and rescue gates. The result replicated on an untouched cylinder topology. We therefore reject the registered claim that these constraints induce a reachability-aligned hidden deformation and reject the downstream geometry-to-behavior causal chain. The result does not show that constraints can never shape representation. It shows, decisively in this regime, that perfect constraint-dependent behavior does not require the proposed geometry and that a constraint alone is not a sufficient causal primitive.

## 1. Question and contribution

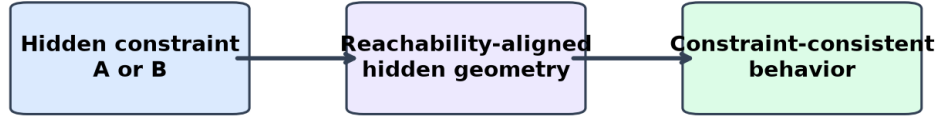
Task context, reward contingency, transition structure, and reachable futures can all correlate with learned representations. The stronger proposal tested here was causal:

*constraint -> reachability-aligned geometry -> behavior.*

A plot or probe cannot establish this chain. The geometry must follow an independently specified future object after a constraint swap, and a state-independent intervention must selectively reverse or impose both geometry and behavior while leaving nuisance competence intact.

The contribution is a controlled counterexample. The agents inferred and used the active constraint perfectly, but the registered future-language geometry did not become the active hidden geometry. The measurement could recover known injected geometry, so the negative was not an insensitive metric null. The targeted transports also failed against severe matched controls. These facts jointly reject the proposed mechanism for the registered model and tasks.

### Registered chain: both arrows had to pass



**Observed: behavior = 1.000, but active reachability geometry and targeted transport both failed**

*Registered causal chain and decision*

## 2. Relation to prior work

Representational similarity analysis formalized comparisons between model and neural dissimilarity matrices [1]. The successor representation established that state similarity can be organized by policy-conditioned future occupancy [2], while reachability-aware representation learning showed that ordinary latent Euclidean distance need not match true control reachability [3].

Context-sensitive RNNs and multitask networks can form task-dependent, orthogonal, or reusable dynamical structures [4-6]. In a hippocampal network model, task-specific latent structure and policy co-evolved; perturbations linked representational biases to behavior [7]. Interchange interventions provide a formal route from neural variables to high-level causal variables [8], and recent manifold steering work directly tests whether representation-respecting edits steer behavior [9].

Those results make constraint-induced geometry plausible, not necessary. Successor features are policy-conditioned; task manifolds may reflect context, readout, value, confidence, or training history; and a fitted latent transform can carry these variables while creating off-manifold states. Empirical-design work further warns that trajectories are not independent replicates and that seed-level uncertainty is essential [10]. Our design therefore used policy-free future languages, paired seed inference, counterbalanced no-swap controls, an injected measurement positive control, and matched latent nulls.

## 3. Exact experimental world

### 3.1 Decision units and hidden constraints

The physical state space was a 6 by 6 torus. A decision unit contained current cell  $x$  and goal  $g$ . The behavior alphabet was accept/reject. The hidden constraint was inferred from demonstrations:

$$C\_A(x, g) = 1[(x\_1 + x\_2) \bmod 2 = (g\_1 + g\_2) \bmod 2]$$

$$C\_B(x, g) = 1[x\_1 \bmod 2 = g\_1 \bmod 2]$$

$$C\_D(x, g) = 1[x\_2 \bmod 2 = g\_2 \bmod 2]$$

$C\_A$  and  $C\_B$  were the confirmatory constraints.  $C\_D$  was an equally balanced, learnable deterministic control. Each rule contained exactly 648 accept and 648 reject units. The same observation alphabet exposed all candidate parity relations but never identified which rule was active. Each demonstration used the same feedback channel, timing, and scalar magnitude. The experiment does not claim "the same information": different rules necessarily carry different conditional information.

### 3.2 Policy-free future language

The independent geometric target was not the model's policy. For physical suffixes  $\alpha$  of length  $H=4$  from {north, south, east, west, stay}, define

```

phi_C,g(u)[alpha] =
  |A_0|^( -H/2)
  * 1[C_C(u)=1 and f_g^alpha(x)=goal].

d_reach,C,g(u,v)^2 = ||phi_C,g(u) - phi_C,g(v)||_2^2.

V_C,g(u) = sum_alpha phi_C,g(u)[alpha]^2.

```

The volume  $V$  was residualized separately. Thus alignment could not pass merely because two units had similar counts of successful suffixes.

### 3.3 Recurrent representation

A recurrent state is history-dependent. We therefore measured

```

eta_t = (o_0:t, a*_0:t-1, r_0:t-1)

h_theta(u,eta_t) = GRU_theta(eta_t, o(u)).

```

The one-layer 48-unit GRU was trained on identical observation schedules for A, B, and D. Query activations were taken after 12 demonstrations, at the GRU output immediately before a fixed linear action head. Eight matched histories were collected for each of 96 balanced decision units.

## 4. Geometry and causal estimands

### 4.1 Cross-validated hidden distance

Repeated histories were split. With split means  $\mu$  and a shrinkage residual covariance  $\Sigma$ , the registered hidden dissimilarity was

```

d_H(i,j) =
  (mu_i^(1) - mu_j^(1))^T
  Sigma^(-1)
  (mu_i^(2) - mu_j^(2)).

```

This crossnobis estimator may be negative in finite samples; no metric nonnegativity claim is made for the estimator.

For vectorized upper-triangle RDMs  $y$  and  $x_C$ , nuisance matrix  $Z$  included sensory distance, unconstrained physical distance, both reachability-volume differences, and all three oracle-action equality RDMs. Matched history identity and probe frequency were exact zero-variance integrity controls. With  $M_Z = I - Z Z^+ +$ :

```

A_C = corr(M_Z y, M_Z x_C)

g = A_B - A_A.

```

The swap estimand used a counterfactual no-swap difference:

```

tau_A->B =
  [g_post - g_pre]_(A->B)
  - [g_post - g_pre]_(A->A).

```

B-to-A was symmetric and had to pass independently.

### 4.2 Low-rank interventions

Decision units were hash-split into 60% transport anchors and 40% untouched tests. Only torus anchors fitted a state-independent affine delta map:

```

T_S->T(h) = h + b + h U V^T

rank(U V^T) <= 4.

```

The map was applied once at the query commitment surface before the unchanged action head. Primary intervention effects were:

$$N\_B = E[\text{Acc}_B(\text{identity})] - E[\text{Acc}_B(T\_B \rightarrow A)]$$

$$S\_B = E[\text{Acc}_B(T\_early \rightarrow B)] - E[\text{Acc}_B(T\_matched\_random)]$$

Targeted effects had to exceed the strongest anchor-permuted or rank-, norm-, and covariance-matched random control, move the registered geometry in the intended direction, preserve linear state/goal decoding within 0.03, keep activation norm and covariance drift below 10%, and show a monotone dose response. "Necessity" and "sufficiency" are restricted to this subspace, surface, and frozen network.

## 5. Preregistered design and gates

Seeds 0-31 were confirmatory. Seeds 1000-1003 were permanently reserved for implementation smoke tests. Hyperparameters, layer, metric, transport rank, probe split, topologies, bootstrap, and thresholds were frozen before confirmatory execution. Seed was the inferential unit. Ten thousand paired bootstrap resamples produced the intervals.

The initial vertical-versus-horizontal fiber design was rejected before model training because its future RDMs were automorphic under action relabeling and its behavior-action marginals were unmatched. That rejected alternative is preserved in the preregistration amendment.

| Gate          | Required evidence                                                                                                     | Result |
|---------------|-----------------------------------------------------------------------------------------------------------------------|--------|
| F0 integrity  | exact schedules, matched marginals, non-collinear RDMs, full-rank nuisance design, VIF < 10, disjoint transport split | PASS   |
| F1 competence | A/B >= .85; 28/32 seeds > .80; D >= .75; sham <= .60; injected lift lower bound > .20                                 | PASS   |
| G1 geometry   | both lower bounds > .05; 28/32 positive; all comparators beaten                                                       | FAIL   |
| G2 swap       | both swap lower bounds > .05; no-swap upper 90% bound < .05                                                           | FAIL   |
| G3 impairment | both directions selectively impaired with nuisance preservation                                                       | FAIL   |
| G4 rescue     | both directions selectively rescued with nuisance preservation                                                        | FAIL   |
| G5 topology   | F1-G4 transport without refitting on untouched cylinder                                                               | FAIL   |

The overall claim was an intersection-union decision. Downstream evidence could not compensate for G1.

## 6. Results

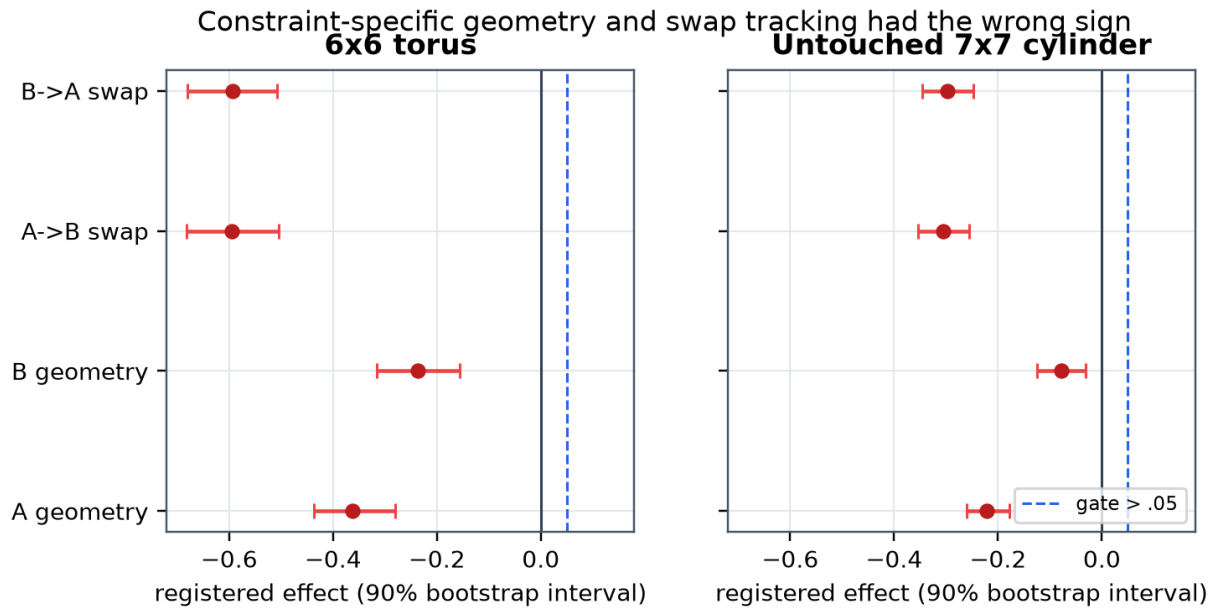
### 6.1 The test was competent and sensitive

All 32 agents achieved 1.000 mean accuracy on A, B, and D on the torus and the untouched cylinder. Every seed exceeded the registered competence threshold. The randomized-label sham was 0.509 (90% bootstrap interval [0.500, 0.518]). The injected-geometry recovery lift was 0.907 [0.864, 0.947] on the torus and 0.559 [0.535, 0.583] on the untouched 7-by-7 cylinder, using the primary calibration precision without refitting. F0 and F1 passed.

### 6.2 Active reachability geometry was absent

The central prediction had the wrong sign. Active-minus-inactive alignment was -0.363 [-0.436, -0.281] under A and -0.237 [-0.315, -0.157] under B. Only 5/32 A seeds and 7/32 B seeds had the predicted positive direction, far below the registered 28/32 requirement. The cylinder replicated the negative direction: -0.222 [-0.261, -0.178] and -0.079 [-0.125, -0.032]. Because competence and measurement

sensitivity passed, G1 is a valid rejection rather than an inconclusive null.



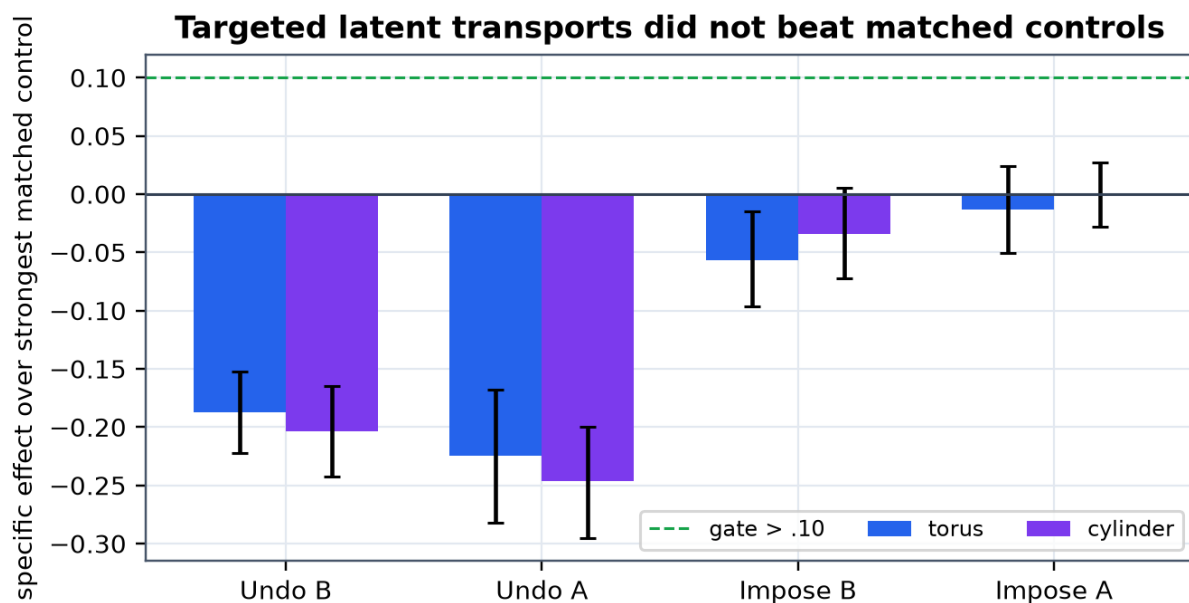
*Constraint-specific geometry and swap effects*

### 6.3 Swaps moved opposite the prediction

The no-swap-adjusted A-to-B effect was -0.594 [-0.682, -0.504]; B-to-A was -0.593 [-0.680, -0.508]. Transfer effects were also negative. The no-swap drift upper 90% bound was 0.067, above the 0.05 tolerance. Thus geometry did not track the active constraint on the registered clock.

### 6.4 Targeted transports did not beat matched controls

Selective impairment effects were -0.187 for undo-B and -0.225 for undo-A. Selective rescue effects were -0.057 and -0.014. A negative specificity value means a matched control changed active behavior at least as strongly as the targeted transport. The transports also violated nuisance-preservation bounds; they were not valid selective mediator interventions. G3 and G4 failed independently of the upstream G1 failure.



| Noncompensatory verdict: upstream geometry failed |      |
|---------------------------------------------------|------|
| F0_integrity_identifiability                      | PASS |
| F1_competence_measurement_sensitivity             | PASS |
| G1_constraint_specific_geometry                   | FAIL |
| G2_swap_tracking                                  | FAIL |
| G3_selective_impairment                           | FAIL |
| G4_selective_rescue                               | FAIL |
| G5_topology_transport                             | FAIL |

Noncompensatory gate verdicts

## 7. What the result disproves

The registered proposition was:

*switching the active successful-future constraint makes hidden geometry more similar to that constraint's policy-free future language, and a targeted low-rank transport of the deformation selectively changes behavior.*

That proposition is false for the tested meta-GRU, constraint pair, intervention class, seeds, and topologies. The stronger informal statement that a task constraint is sufficient to make reachability geometry the central causal primitive is therefore disproved by this counterexample.

Three weaker claims survive:

1. Agents can infer and use hidden future-admissibility rules from matched demonstrations.
2. Their recurrent states change with context and history.
3. A known geometric signal would have been detectable by the registered analysis.

None implies that the model organizes pairwise similarity by the complete successful-future language.

## 8. Why behavior can succeed without the proposed geometry

The observation exposed all candidate parity relations. A recurrent context state only needed to select which relation fed the fixed binary readout. That computation can be implemented by gating or directional readout without making the complete pairwise hidden RDM resemble the future-language RDM. Distance geometry also cannot detect codes that differ only by transformations preserving pairwise distances.

This yields the main correction:

*Constraint-dependent behavior does not entail constraint-shaped global distance geometry.*

Future work should distinguish at least four objects: feasibility language, policy-conditioned occupancy, decision boundary, and recurrent dynamics. Whichever object is claimed causal must be the object directly intervened on.

## 9. Contradictions and reliability audit

The practitioner lens says perfect behavior is the operative fact; the model found a cheaper relation-selection solution than enumerating future languages. The academic lens says task geometry and successor-like representations are real but conditional phenomena. The skeptic notes that any one

RDM family is incomplete; our conclusion is therefore metric-specific. The incentives lens warns that positive manifold stories are easier to publish than severe nulls. The historical lens places this result in the recurring correction from decodability to causal use.

The evidence ranking is:

1. **High reliability:** exact finite enumeration, matched marginals, complete seed set, competence, sham, and injected positive control.
2. **High reliability within metric:** the sign-reversed G1 and G2 estimates.
3. **Moderate reliability:** transport failure, because the transports violated nuisance-preservation bounds as well as specificity.
4. **Not established:** absence of all task-relevant codes, other metrics, persistent recurrent interventions, other architectures, or a universal claim about meaning and goals.

## 10. Reproducibility and provenance

The public package contains the frozen preregistration and manifest, exact world enumerator, training code, seed-level public rows, gate logic, figure builder, manuscript, and deterministic PDF builder. Raw checkpoints and the complete run payload remain in gitignored artifacts/.

Registered execution:

```
uv run --no-sync python -m \
  experiments.constraint_swap_causal_geometry.run_experiment \
  --seeds 32 --workers 4
```

The seed rows are sufficient to recompute every interval and gate. The paper reports negative and rejected artifacts rather than deleting them.

## 11. Conclusion

The experiment decisively rejects its scoped causal-deformation hypothesis. Constraints A and B completely controlled successful behavior, but neither became the active policy-free future-language geometry. Geometry failed before causal intervention was considered, swaps moved opposite prediction, and targeted transports did not beat matched nulls. The broad research program remains open, but its principle must be weakened:

*A constraint can shape representation under some objectives and architectures; it does not, by itself, force a reachability-aligned geometry or make that geometry the causal basis of behavior.*

## References

1. Kriegeskorte N, Mur M, Bandettini PA. Representational similarity analysis - connecting the branches of systems neuroscience. *Frontiers in Systems Neuroscience*. 2008;2:4. doi:10.3389/neuro.06.004.2008.
2. Dayan P. Improving generalization for temporal difference learning: the successor representation. *Neural Computation*. 1993;5(4):613-624. doi:10.1162/neco.1993.5.4.613.
3. Wang K, Zhou K, Feng J, Hooi B, Wang X. Reachability-aware Laplacian representation in reinforcement learning. *ICML/PMLR* 202. 2023.
4. Mante V, Sussillo D, Shenoy KV, Newsome WT. Context-dependent computation by recurrent dynamics in prefrontal cortex. *Nature*. 2013;503:78-84. doi:10.1038/nature12742.
5. Flesch T, Juechems K, Dumbalska T, Saxe A, Summerfield C. Orthogonal representations for robust context-dependent task performance in brains and neural networks. *Neuron*. 2022;110:1258-1270.e11. doi:10.1016/j.neuron.2022.01.005.
6. Driscoll LN, Shenoy K, Sussillo D. Flexible multitask computation in recurrent networks utilizes shared dynamical motifs. *Nature Neuroscience*. 2024;27:1349-1363. doi:10.1038/s41593-024-01668-6.

7. Cone I, Clopath C. Latent representations in hippocampal network model co-evolve with behavioral exploration of task structure. *Nature Communications*. 2024;15:687. doi:10.1038/s41467-024-44871-6.
8. Geiger A, Lu H, Icard T, Potts C. Causal abstractions of neural networks. *NeurIPS*. 2021.
9. Wurgaft D et al. Manifold steering reveals the shared geometry of neural network representation and behavior. *arXiv:2605.05115*. 2026. Preprint.
10. Agarwal R, Schwarzer M, Castro PS, Courville AC, Bellemare M. Deep reinforcement learning at the edge of the statistical precipice. *NeurIPS*. 2021.